

Virtual memory

Module V, Part 1

Introduction

- Virtual memory is a memory management technique that **creates an illusion for applications that they have more RAM than is physically available.**
- It achieves this by **temporarily moving inactive data from RAM to disk storage** (paging/swapping), allowing computers to run larger applications, enhance multitasking, and improve overall system efficiency.
- OS does this mapping

Introduction

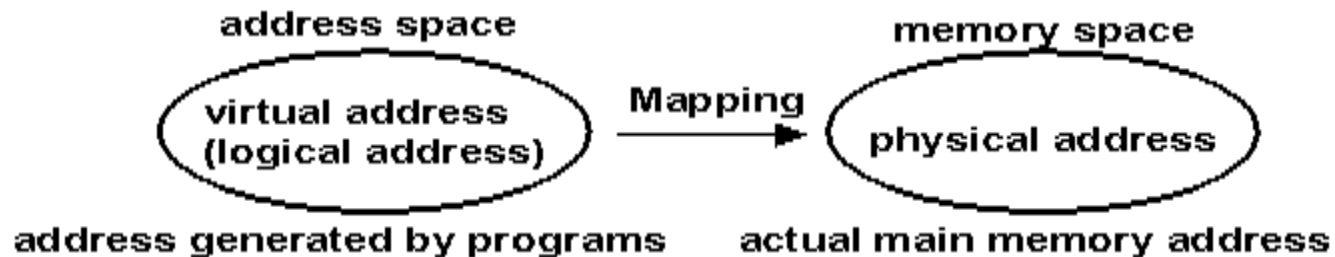
- Programs and data are stored in auxiliary memory.
- Portions of a program or data are brought into main memory as and when it needed by the CPU.
- Virtual memory is a **memory management technique** that appears such that all addressable memory (disk memory) resides in main memory. It provides programmers the illusion that they have a very large memory at their command.
- **Improves Efficiency** by allowing better use of computer resources

Introduction

- Two characteristics fundamental to memory management:
 - 1) all **memory references are logical addresses** that are **dynamically translated into physical addresses at run time**
 - 2) a process may be broken up into a number of pieces that don't need to be contiguously located in main memory during execution
- If these two characteristics are present, it is not necessary that all of the pages or segments of a process be in main memory during execution

Virtual Memory

- It gives the programmer the illusion that the system has a very large memory, even though the computer actually has a relatively small main memory



- It provides a mechanism for translating virtual address to physical address
- Mapping is handled by the H/W automatically by means of a mapping table

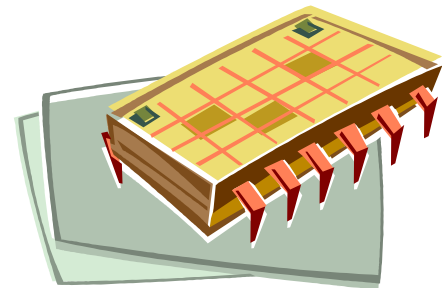
Real and Virtual Memory

Real memory

- main memory, the actual RAM

Virtual memory

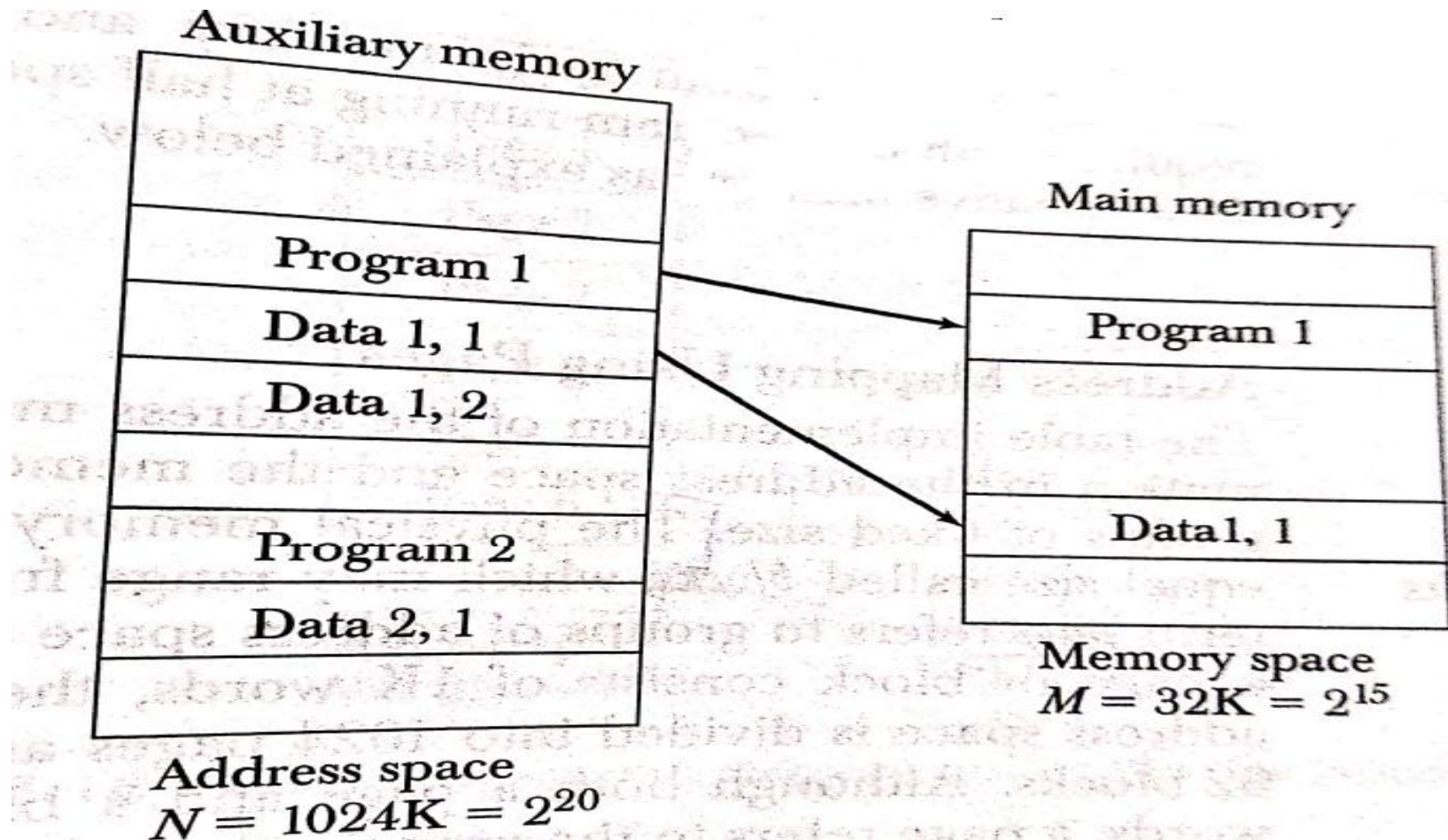
- memory on disk
- allows for effective multiprogramming and relieves the user of tight constraints of main memory



Virtual memory

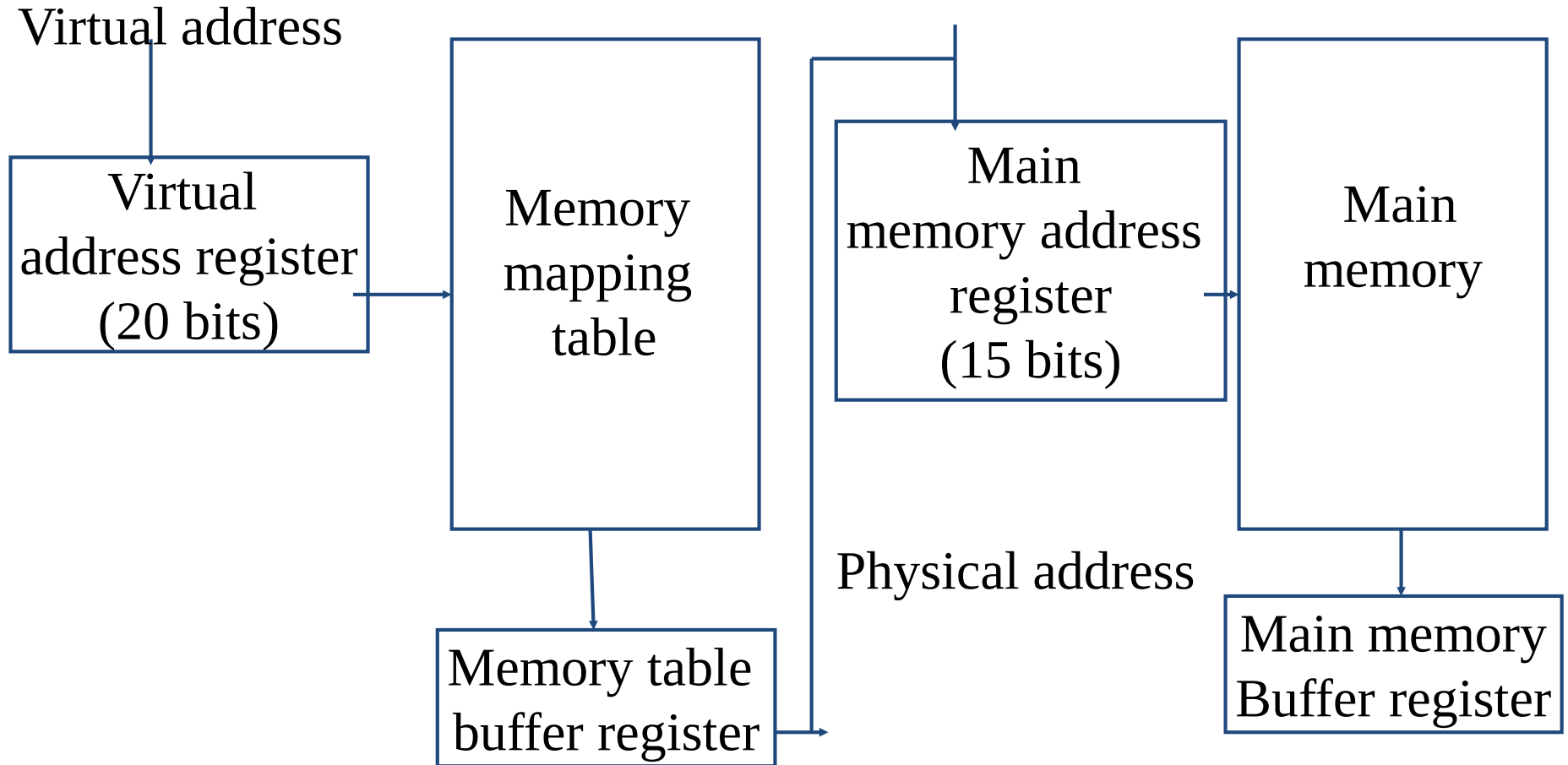
- Address used by a programmer is called as **virtual address** and the set of such addresses is called **address space**.
- An address in main memory is called as **physical address** and set of such locations are called as **memory space**.
- **Address space is allowed to be larger than the memory space** in computers with virtual memory.

Relation between address and memory space in a virtual memory system.



- A computer with **main memory capacity of 32k words** requires 15 bits to specify the physical address in memory. Suppose the computer has available **auxiliary memory for storing $2^{20} = 1024K$ words**, then auxiliary memory has a capacity of storing 32 main memories ($2^{20} / 2^{15} = 2^5$). Denoting **address space** by N and **memory space** by M, we have $N=1024K$ and $M=32K$ words.

MEMORY TABLE FOR MAPPING A VIRTUAL ADDRESS



Address space and memory space are split into fixed size group of words called blocks or pages (Eg.1K words)

Auxiliary
Memory

Page 0
Page 1
Page 2
Page 3
Page 4
Page 5
Page 6
Page 7

Address space
 $N=8K=2^{13}$

Main Memory

Block 0
Block 1
Block 2
Block 3

Memory space
 $M=4K=2^{12}$

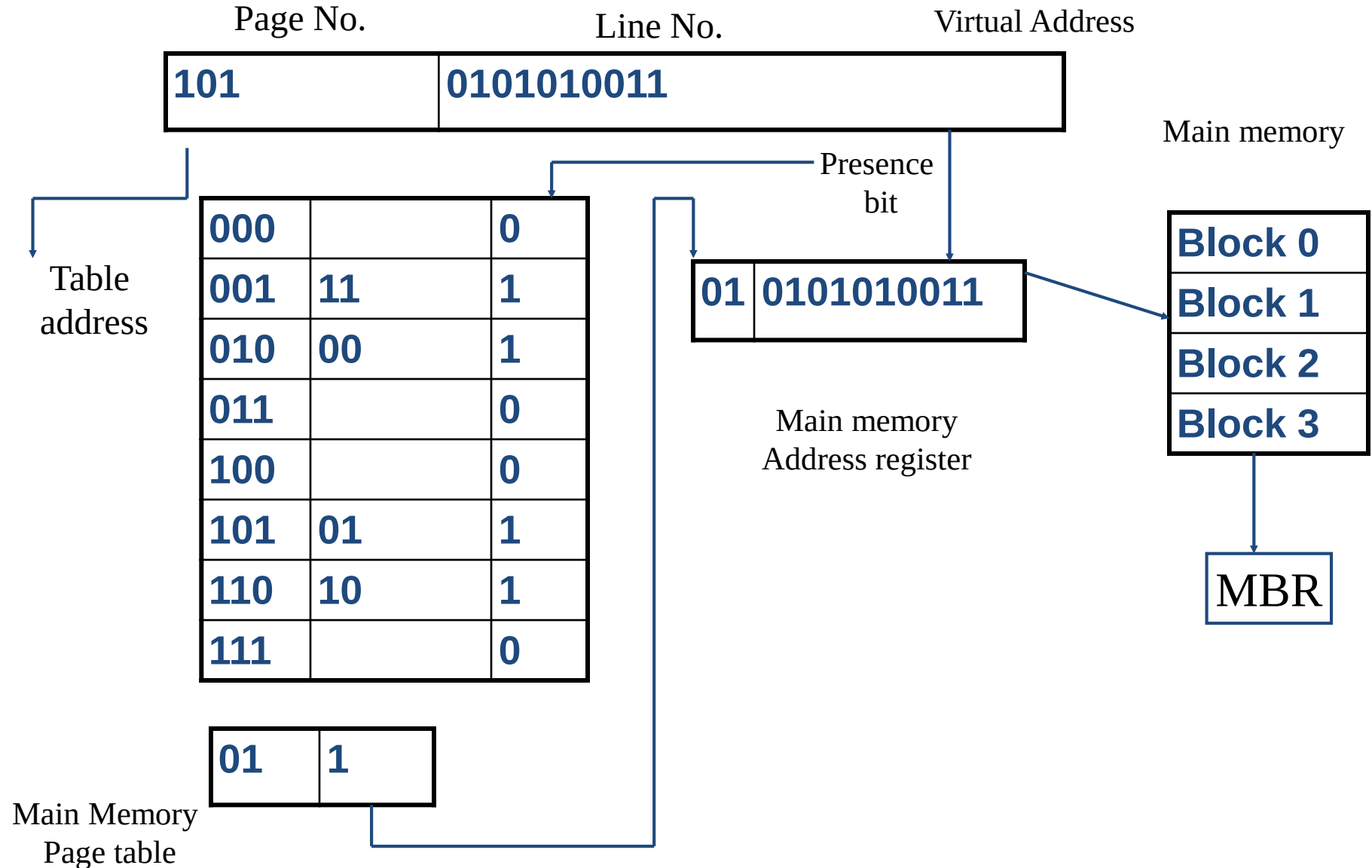
Address Mapping using Pages

- Line address in address space and memory space is the same.
- The only mapping required is from a page to a block.
- Memory page table gives the block of the page
- Presence bit indicates whether the corresponding page is in main memory or not
- If number of blocks in memory = m
- Number of pages in virtual address space = n
- n entry page table in memory is required

PAGING

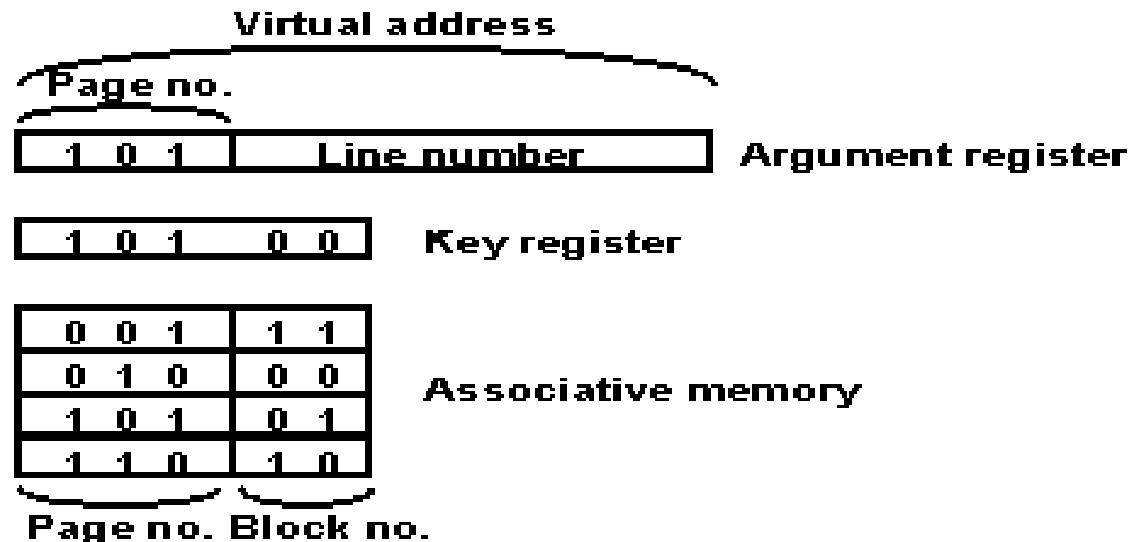
- Split memory into equal sized, small chunks called **page frames**
- Split programs (processes) into equal sized small chunks - pages
- Allocate the required number of page frames to a process.
Each process has its own page table
- Operating System maintains list of free frames
- Use page table to keep track
- Each page table entry contains the **frame number of the corresponding page** in main memory
- **A bit is needed to indicate** whether the page is in main memory or not

Organization of Memory Table in a Paged System



Associative Memory Page Table

- Here number of words in page table = number of blocks in main memory
- Implemented using associative memory with each word in memory containing a page number together with its corresponding block number
- Page fault – page number cannot be found in page table



- A virtual memory system is a combination of **hardware and software** techniques
- H/W mapping mechanism and Memory management software together forms the **architecture of virtual memory**
- Memory management software
 - Which *page to be removed to create a room for a new page*
 - When *new page is to be transferred from secondary memory to main memory*
 - *Where the page is to be placed in main memory*
- Loading a page from secondary to main memory is an I/O operation. So when **page fault occurs**, CPU suspends the **present program** and control is transferred to next program which is waiting for CPU.

MODIFY BIT IN PAGE TABLE

- **Modify bit** is needed to indicate **if the page has been altered since it was last loaded into main memory**
- If no change has been made, the **page does not have to be written to the disk** when it needs to be swapped out

Eg. An address space is specified by 24 bits and corresponding memory space by 16 bits.

- How many words are there in the address space?
- How many words are there in the memory space?
- If a page/block consists of 2K words, how many pages and blocks are there in the system?

Address space—24 bits

Memory space—16 bits

a) Words in address space — $2^{24} = 16\text{M words}$

b) Words in memory space -- $2^{16} = 64\text{Kwords}$

c) Page size – 2K words

How many pages --- $2^{24} / 2^{11} = 2^{13} = 8\text{K Pages}$

How many blocks --- $2^{16} / 2^{11} = 2^5 = 32 \text{ blocks}$

1. The logical address space in a computer system consists of 128 segments . Each segment has up to 32 pages of 4k words in each. Physical memory consists of 4k blocks of 4 words in each. Formulate logical and physical address formats

logical address

7 bits	5 bits	12 bits = 24 bits
Segment	page	word

physical address

12 bits	2 bits
Block	word